Module Title:	Computer Systems Security 2026
Module Code:	B9IS103
Module Lecturer:	Paul Laird
Assessment Title:	Secure Communications/Collaboration System Design and Deployment Analysis of Secure System Deployment
Assessment Number (if relevant):	1
Assessment Type:	Practical (lab-based)
Restrictions on Time/Length:	N/A
Individual/Group:	Group
Assessment Weighting:	100%
Planned Feedback/Results Release Date:	In class; Written within four weeks of submission
Mode of Submission:	On-line **ONLY Moodle**

The assignment is a group assignment.

Requirements: You are required to submit an application and accompanying report, which may be written in Python or any other suitable programming language. Please comment code and use appropriate variable names. You **MUST** make use of a publicly hosted git repository (e.g. GitHub) to **DEVELOP** (not upload).

Submission: Zip the Solution and Project folder and upload using the CA_ONE link on Moodle. Include links to your code repository and deployment location in the online text section.

IMPORTANT NOTE: Only one submission is allowed so please ensure that you have everything that is required in the zip before uploading to Moodle.

The following table illustrates the percentage allocation for each individual part of the assignment. Grades will be awarded on the basis of the documented artefacts or attacks delivered, and their presentation in class.

Total Marks (100)

Problem	**Detail**	**Breakdown of Marks**
Design and Implementation	Design specification plus relevant Code	40 marks
Vulnerability Analysis	Analysis of potential vectors of attack on two other systems	20 marks * 2
Reactive analysis and improvements	Response to, fix of, or planned mitigation of attacks illustrated	20 marks

The task:
Design and deploy to the cloud an encrypted communications/collaboration application, which must, at a minimum, allow for messages to be encrypted between two parties who have not met to exchange keys.
You can use a self-hosted server as a relay, or you can integrate into a social medium or email for communications.

Some form of authentication of the identity of the participants should be used, in line with the form of identity selected. The relevant service providers may be assumed to provide their services in a correct and trustworthy manner in terms of execution, but MUST NOT be trusted not to leak any confidential information (They must be considered maliciously curious). Examples include, but are not limited to:

- PKI/DNS based identity, with keying information posted on https sites with valid certificates
- Social media based identity, with x (twitter) / facebook / other channels / profiles being used to certify identity
- Communications channel based identity, e.g. email and/or SMS
- Idiosyncratic identity verification (could include authenticated media transmission during setup phase, or novel ideas)
- Combinations of the above

You must outline the requirements of the system, including functional and nonfunctional requirements, and specifically detail the security requirements.

The system must not rely upon any secret algorithm or security through obscurity, although you may use (only as an additional layer) steganography for plausible deniability. Your system must be documented in sufficient detail so as to allow another developer to efficiently augment your work, or thoroughly test its security (README.md or Word online).

Every system has weaknesses, and can at very least be attacked on the basis of the assumptions upon which it relies. You will be assigned two other groups' projects to analyse, and must describe how you would propose to attack the system if you were attempting to intercept relevant communications, and also describe any other potential vectors of attack or weakness.

Finally, you should attempt to rebut or to outline mitigation strategies for any attacks revealed by other groups.

Please note that any resources or assistance from stock solutions through guides, tutorials, AI (Claude, Gemini, ChatGPT etc.), human or any other assistance, are permitted inasmuch as they are documented and attributed (see below, and ChatGPT or other LLM logs and **prompts must be retained**, as well as any links to resources and **ANY communication with friends who help**). It must be clear what was the contribution of each group member and what was from sources mentioned above **Anything from any external source must be added in its own single commit and not mixed with students' work or spread over multiple commits**. Detailed questions may be asked at presentation time, and each student's contribution must be sufficient to demonstrate achievement of the learning outcomes.

Resources

Software or tools required/ useful: Azure, GitHub.dev, GitHub spaces
Templates, files, data tables provided by the lecturer: Moodle and GitHub resources

Generative Artificial Intelligence Assessment Scale

Can generative AI be utilised in this assignment?

1	2	3	4	5
NO AI	**AI-ASSISTED IDEA GENERATION AND STRUCTURING**	**AI-ASSISTED EDITING**	**AI TASK COMPLETION, HUMAN EVALUATION**	**FULL AI**
The assessment is completed entirely without AI assistance. This level ensures that students rely	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. **No AI content is**	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. **You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.**	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. **You may use AI**

solely on their knowledge, understanding, and skills. **AI must not be used at any point during the assessment.**	**allowed in the final submission.**	be created using AI. **AI can be used, but your original work with no AI content must be provided in an appendix.**		**throughout your assessment to support your own work and do not have to specify which content is AI generated**
			The use of AI-generated content is treated in the same manner as the use of any other external resource such as libraries, packages or frameworks. It must be clearly identified and attributed, and credit will be limited to the student's own work, to the exclusion of such resources, except for how students integrate and deploy all of these resources. Given the nature of the assignment, it is expected that AI generated content will feature heavily in the submissions, but must be understood and declared. Questions will be asked during the presentation.	